

Institute of Actuaries of India

Subject ST2 – Life Insurance

March 2018 Examination

INDICATIVE SOLUTION

Solution 1:

- i) Following needs of the customers can be met by a with-profit endowment product:
- i. To save money for retirement
 - ii. To repay the capital on interest-only loan
 - iii. To transfer wealth, say from parents to children
 - iv. To provide financial protection to dependents in case of death of the life assured
 - v. To save money for future expenses like buying house, car or to fund child education
- [4]
- ii) Following are the risks that the Company will expose itself to:
- i. Business risk: the Company has so far been selling unit linked products only. Moving from unit linked business to with-profit business will require a lot of sales training, change in product placement strategy and all this may take time. This can adversely impact the business / sales during the transition phase
 - ii. Since the Company has been selling only plain vanilla unit linked products, it was able to pass the entire investment risk to the policyholders. However in case of with-profit contracts the company may be able to do so and will also have to share the investment risk.
 - iii. Guarantee risk: In case of plain vanilla unit linked product the benefit amount is directly linked with value of the underlying assets and no guarantees are offered. However in case of with-profit contracts there is underlying guarantee in the form of guaranteed maturity benefit, bonuses already declared. Hence the Company will be required to honor such guarantees irrespective of the performance of the underlying assets and may be required to hold extra capital to meet such guarantees
 - iv. Due to guarantee nature of with-profit contracts, the company may not want to invest all its fund in equity. This may impact the expected future return and may not go well with the policyholders who were expecting return similar to existing ulip products
 - v. Operational Risk: The Company has so far has not done any business in with-profit category. With-profit category of business has its own set of requirements, like regular calculation and monitoring of asset share value, bonus declaration, etc.. Any failure due to system or people may adversely affect the company
 - vi. Mis-selling: The guarantee nature of with-profit can be mis-placed by the sales person and hence may lead to mis selling. Also since with-profit contracts are relatively complex in nature, there is a risk that the Customer has not properly understood the product and this may create customer grievances.
 - vii. Persistency risk: At times when the asset share is negative there is financial risk from withdrawal. Also if the guaranteed surrender value is greater than the asset share, there is financial loss to the company from early termination of the contracts.
 - viii. Closure of unit linked products will also lead to higher lapses
 - ix. Higher lapses will lead to lesser contribution to overhead expenses thereby resulting in lower expense loading and expense risk
 - x. Closure of unit linked business might also lead to potential reputation and competition risk, hence lower new business volume for the Company

- xi. There could be operational issues if the unit linked business is closed completely. Unit Linked business staff may require to be retrained to with-profit business or the Company may be required to lay-off such staff.
- xii. Shift from market practice: The change in strategy will also depend on what the competitors are doing and what the customers want. If this strategy is not aligned with the competition and customer's need, then this will adversely impact the company in terms of lost business opportunity, brand name, etc.

[8]

iii) (i) Insurer Point of View:

1. Offering guarantee may act as a better sales proposition and can be attractive to the customers, thereby increasing the sales for the company
2. But if the sale is too high, it might cause new business risk
3. With profits business implies investing in equity and property which is volatile. Since with-profit products have guaranteed benefits in them, exposing the fund to high risky assets can lead to greater investment risk
4. buildup of guarantee in the form of higher guaranteed addition will lead to:
 - a. increasing the investment risk and capital requirement
 - b. threaten the overall solvency of the company.
5. The company has not offered such a guaranteed addition before, so it may be not be able to value the guarantee correctly...

.....leading to either the guarantee not being value for money for the customer, or losses for the company
6. May result in restricting the investment freedom, and hence potential investment growth
7. In times of favorable market condition since there is no RB in the product the Company will have to stick with GA of 2% only and will not able to increase it...

....this might lead to selective withdrawals and reputational risk
8. Since the product does not have any concept of RB, it will not be able to adjust the benefit amount if the asset share falls below the guaranteed benefit (sum assured) plus guaranteed addition, thereby leading to:
 - A. depletion of estate (free assets) quickly, thereby eliminating any investment freedom that the company may have
 - B. Not declaring any terminal bonus. This might lead to....
 - C. ...PRE (Policyholder Reasonable Expectation) issues if terminal bonus was illustrated to the customers at the point of sales

Policyholder point of view:

1. Guaranteed addition will act as a better value to those customers who don't want to expose themselves to market volatility and who are looking for secured returns
2. Offering Guaranteed Addition in lieu of RB may not be beneficial to customers at times of favorable market conditions.
3. Customer may not like the idea of receiving all the profit at the end of the contract in the form of terminal bonus
4. Higher guarantee in the form of guaranteed addition may lead to higher capital requirement.

In order to meet this higher cost of guarantee and capital the company may decide to levy charge. This will ultimately impact the overall return given to the policyholders....

or it may not help the customer meet their needs

5. If the equity / property market performs poorly, then due to the guarantee nature of the product the Company may have to cut the terminal bonus completely. This may not go well with the policyholders

[11]

- iv) In order to calculate the cost of guarantee arising due to guaranteed benefit, the actuarial student should use stochastic model, as stochastic model allows a probability distribution to be attached to market value and reinvestment yields of various asset classes. [2]

- v) Steps involved in modelling:

1. Since stochastic model involves running of lot of simulations, it is best practice to group the data that we will be examining to reduce the run time of the model
2. Next step will be to choose the probability density function of various asset classes, for e.g. we may select log normal model for projecting equity returns, Hull-White model for debt instruments, etc.
3. Next will be to assign any correlation between various variables involved. The student may look at modelling policyholder behavior while calculating the cost of guarantee
4. Then the student will have to construct a model by considering all the possible cash flows
5. The model then should be able to project these cash flows and hence the asset share of the policy.
6. The asset share of the policy should also allow for deduction with respect to cost of guarantee charge.
7. To begin with the student need to assume one guarantee charge and make deduction to the asset share
8. The projected asset share at maturity for each of the simulation will then need to be compared against the guaranteed benefit offered so as to see if there is any shortfall arising
Shortfall may be defined as = (Guaranteed Benefit – Asset Share at Maturity)
9. Finally the cost of guarantee is defined as present value of shortfall discounted using risk-free rate
10. Since the cost of guarantee depends on the asset share and the asset share itself depends on the guarantee charge, this could lead to circularity effect.
11. Hence the above process needs to be repeated number of times such that the present value of guarantee charge equates with the cost of guarantee, so as to be risk neutral

[7]

- vi) Following cash flows needs to be considered:

- i. Premium amount
- ii. Expenses
- iii. Commission

- iv. Death benefit outgo
- v. Investment income on asset share
- vi. Guaranteed additions of 2%
- vii. Terminal bonuses
- viii. Transfer to shareholders
- ix. Taxes, if any
- x. Cost of guarantee and cost of capital charge

Following assumptions / parameters would be required to make:

- i. Expenses assumption
- ii. Future investment return
- iii. Probability distribution for each asset classes
- iv. Mortality assumption
- v. Corporate tax rate applicable
- vi. Inflation assumption
- vii. Average premium and average sum assured
- viii. Risk free yield curve for discounting the shortfall
- ix. Lapse / surrender rate

[4]

[36 Marks]

Solution 2:

i) In general, the following factors could be analyzed:

- type of contract
- duration in force.
- Distribution channel used
- Target Market
- Frequency of premium payment
- Size of Premium
- Premium payment method;
- Original term of contract
- Gender
- Age

[4]

ii)

- The factors that need to be considered and the order of importance of the factors depends on the purpose of the analysis.
- The factors chosen should be such that there is enough data in each “cell” to warrant separate analysis
- The overall size of the product when compared to the overall portfolio of products offered by the Company
- Whether persistency is a key risk factor for the product (e.g. for annuities it would not be important or possible)

[2]

iii)

- The large capital expenditure will have to be excluded completely from the analysis.
- However, given this is a cost for an ongoing administration system then the costs should be amortized over the expected useful lifetime of the system
- Or, a notional “rent” for the administration system, as if it was being rented from a software company, could be charged
- It could be then be added to overhead costs
- The insurer needs to assess the ongoing maintenance/update costs of the new administration system and these costs need to be included in the expense analysis.
- Allocated to departments in the proportion of usage of the new admin system
- There could be costs associated with decommissioning the old administration system – both ongoing maintenance and one off costs and these should be considered alongside the new administration system costs

[4]

iv) Future Steps for the company:

- Firstly, the company should check the accuracy of the analysis, the data and the models.
- Was the data large enough to be credible?
- What was the basis of the pricing assumption? Was there an error there?
Is the mortality experience out of line from the market?
...If so, then the reason for this needs to be investigated. And corrective action taken.

Have there been changes in the terms of competitor products (e.g. stricter underwriting) – leading to a different class of lives purchasing the product from the Company

- The company needs to assess the profitability of the existing product, in order to decide whether to continue to sell the product.
- If the product is unprofitable, the company either have to withdraw the product for new business or re-price the product or redesign the features of the product or consider appropriate underwriting controls.
- If the product is profitable, the company will have to analyze the sources of profitability and reasons for the difference, given the substantial difference in the mortality assumption in the pricing and the experience analysis.
- If the product is profitable, the company needs to examine whether to continue selling the product with such differential, if so, the company need to be aware about the parameters that could go wrong in the future threatening the profitability of the product.
- The company needs to consider whether mortality is a key factor for this product – it may be that the mortality experience is not a profit driver and hence the company may not be concerned about the experience
- The product might have been sold to a class of lives that is expected to have a different experience from that expected at the time of setting the pricing assumption. In such case, adjustments may need to be made in the pricing or underwriting to reflect the actual target market.
- The above situation might have arisen because of possible change in the target market or distribution channel that was used or may be possibility of excess or meagre exercising of any existing options available in the product. Depending on the reasons, the insurer need

to offer the product to either appropriate target market or through the appropriate distribution channel or re design the product options suitably or adjust the assumptions to reflect the new experience.

- Another possibility could be that the underlying mortality rates of the particular class of lives covered under the product would have changed over time. In such case, the insurer needs to study the mortality trends for that particular class of lives and revisit the mortality assumption.
- There could also be a possibility of lenient or strict underwriting controls applied which might have resulted in a different experience. This may be different from the controls that were assumed within pricing. For this, suitable amendments need to be made to the underwriting controls to reflect the expected class of lives to be covered under the product.

[7]

[17 Marks]**Solution 3:****i) Components of asset share:**

Monies in:

- i. Premium Income
- ii. Investment Income
- iii. Profits from without-profit business, if any
- iv. Surrender profit or losses, if any

Monies out:

- i. Expenses including commission
- ii. Cost of providing benefits in excess of asset share
- iii. Taxes
- iv. Transfer of profits to shareholders
- v. Cost of capital
- vi. Contribution to the free assets

[2]

ii) Various checks could be as follows:

- i. Completeness check to ensure all the relevant cash flows have been taken into account while calculating the asset share
- ii. Check if the expenses are consistent with the financial reported numbers of each historical years and does not look too unreasonable
- iii. To check at what level the asset share has been calculated – model point level or group of policies. If grouping has been done need to check what is the deviation in result due to grouping for some samples
- iv. To check if the historical transfer to shareholders coming out from the model reconciles with actual historical transfers in the financial accounts of the company
- v. Check for commission – to check if the commission loaded into asset share calculation reconciles with actual commission paid and is as per the product design.

- vi. Premiums paid – to check if the premium in the asset share calculation reconciles with the actual premium received. Actual policy data from the policy administration system can be used for this reconciliation
- vii. The actuarial team can develop independent sample model to check to validate the model used for reporting of the numbers
- viii. Check if the tax rate for historical years is consistent with the actual tax rate applicable for that financial year
- ix. Check if the historical mortality rate used in asset share calculation is consistent with actual experience of the company. The company would have historical mortality experience analysis, this can form basis for reconciliation and validation. Accordingly the decrement needs to be set up in the model
- x. Check if the any other source of revenue like non-participating profit, if allowed as per company policy, has been taken into account while arriving at the asset share numbers
- xi. If the bonus philosophy of the company is to distribute surrender surplus to the existing policyholders, then such cash flows also needs to be allowed for in asset share calculation
- xii. Count of policies remaining in the model at the end of the financial year can be reconciled against actual data from the policy admin system to check if the decrement chain is working correctly

[8]

iii) Reasons why value of free assets could be reducing are as below:

- i. Error in model / cash flows: This can be due to incorrect assumption used in the calculation of asset share, for e.g. higher than actual expenses taken into account while calculating the asset share
- ii. High level of guarantee: The with-profit product offered by the Company may have high level of guarantee built in them. The level of guarantee might have exceeded the value of assets due to which the company was forced to pay more than the asset share to the policyholder, thereby leading to depletion of estate (free assets)
- iii. This could also be due to smoothing of bonus rate in the past. The company may not have adequately responded to change in economic condition in the past by reducing the level of bonus, as a result of which it had to pay more than the earned asset share of the policy
- iv. Losses on non-participating category of business, if such profits / losses are shared with with-profit policyholders. The company might have missed debiting the asset share with such losses thereby resulting in paying more than the earned share and hence depletion of free assets
- v. Failure to impose Market Value Adjustment on surrenders, might result in paying more on surrender than the earned asset share of the policy
- vi. Higher actual expenses incurred than what is loaded in asset share will result in early depletion of estate. Due to reason of PRE and with-profit philosophy the Company might have imposed a cap of the level of expenses that can be charged to asset share.

In case the actual expenses exceed this cap then the excess will need to be funded by the estate (free asset) of the with-profit fund.

- vii. Policyholder Reasonable Expectation (PRE): Failure to reduce the regular bonus or to cut the terminal bonus altogether during unfavorable market conditions due to PRE concerns may result in depletion of free assets
- viii. Assets and liabilities may be mismatched – leading to a loss in free assets.

[8]

[18 Marks]**Solution 4:**

i)

- The roll out of universal term insurance plan could have significant impact on the company if the premium rates offered under the existing term plan are uncompetitive compared to the universal term plan.
- In particular, with regard to the existing policies where the sum assured is equal to or less than the fixed sum assured of the universal term plan:
 - The company can expect huge lapses on these policies and re-entry through the universal term plan.
 - This would possibly impact the expected premium income from these policies.
 - There would be less number of policies to spread the cost of expenses already incurred.
 - As it is term plan, the asset shares may be very low or nil, resulting in a strain.
 - In turn, impacts the profitability of the product portfolio overall.
- Conversely the Universal Term Plan may prove very popular, increasing new business volumes and thus the number of policies over which expenses may be spread
- The company may also experience lapses from the existing policies where the sum assured is above the fixed sum assured. This could happen, if the existing policyholders intend to take advantage of the competitive rates under the universal term plan. In such cases, this could also result in lapse and reentry:
 - Taking the universal plan for the fixed sum assured and /or;
 - Taking out the balance sum assured under the existing term plan or may not take any further term plan;
- Class of lives expected to have higher mortality may opt to retain with the company in case if they are covered for higher sum assured than the fixed sum assured under the universal term plan. This would lead to selection against the company and may mean the retention of the “bad” risks.
- Pricing as well as reserving assumptions may need to be changed to reflect all of the above
- The company may reconsider continuing to sell the product.
- Overall the impact on the portfolio is, loss of expected premium income from the portfolio, higher than expected expenses to be recovered, possible higher claims due to retention of bad risks, negative impact on the profitability of the product.

[8]

ii)

Stop Loss

- + May help manage total portfolio loss
- + if the fixed sum assured is not very large, rather than reinsuring per life, this might work better in managing the risk
- + could be better at managing aggregation risk
- Since the company will have to mandatorily accept risk, the company will not be able to restrict aggregation risk
 - It may be more expensive
 - Could be administratively difficult
 - Would require annual renewal, so the terms or availability for future years is not guaranteed
 - Company would still be exposed to volatility in claims experience up to the retention level..
 - ... however, if the terms of the universal Term Plan are attractive, there might be large portfolio of lives insured.....
 - ... by law of large numbers, this might be sufficient in reducing the claims volatility.

Proportional Reinsurance

- + Could be possible to negotiate rates that are guaranteed for the entire term
- + even if the rates are reviewable, a treaty could ensure that the existing business is reinsured for the term of the policy
- + Might help with managing risk exposure, particularly if the new business is higher than expected, and goes beyond the risk appetite of the insurer
- + Would dampen the impact of claim volatility
- + Would be administratively simpler
 - May not be effective in covering aggregation risk
 - If the fixed sum insured is very small, could lead to very small amounts being reinsured per life, making the administrative cumbersome

[7]

iii)

Factors that need to be considered before deciding on the reinsurance structure include:

- The company needs to analyze the kind of risks it is exposed to under the product.
- In this particular case, the insurer is entering into a market where the insurer has no control either on pricing or underwriting.
- The insurer may also experience completely different experience compared to the existing term plan.
- The company is also unaware about the profitability of the product and the possible new business under the universal term plan.
- In managing and mitigating such risks, the company need to consider:
 - whether the exposure requires the use of reinsurance.
 - If it requires, the type(s) of reinsurance that are appropriate to use.

- If the relevant reinsurance protection is available, how much reinsurance to use.
 - The reinsurance rates being offered by reinsurers
 - The financial position of any reinsurers in the market
 - Whether the company can afford not to reinsure – it may decide that the loss of profit from the UTP will not make the reinsurance worthwhile
- The company may need to examine whether it can manage the portfolio through the use of mortality fluctuation reserve, given that the sum assured is fixed to all the policyholders. For this the company has to evaluate the cost of financing an appropriate mortality fluctuation reserve need to be analyzed vis-à-vis the cost of obtaining the reinsurance
 - If reinsurance protection has to be opted for, the company may examine whether the existing reinsurance program, if any, with regard to the existing term plan can be extended to the universal term plan.
 - And if extended, whether any additional support such as catastrophe reinsurance, stop loss reinsurance or financial reinsurance need to be considered given that the company may get exposed to huge accumulation risk across the country and also may be subjected to substantial new business strain, if large number of policies are sold.
 - The company may also consider the possibility of getting technical support in terms of managing the business from the universal term plan in the initial period.
 - If the company has to enter into a reinsurance agreement afresh, the insurer, in the process of setting, the Retention limits need to evaluate the company's risk appetite, the reinsurance terms available for such product types, its experience in underwriting existing term plan, any regulatory requirements, any profit-sharing arrangements with the reinsurer, company's ability absorb losses in terms of free assets, reinsurance arrangements in similar products.

[8]

[23 Marks]**Solution 5:**

i)

- Making the policy paid-up
- change the term of an assurance.
- alter the sum assured i.e. increase or decrease.
- alter the premium payable
- Converting one type of policy to another type of policy.
- Taking a premium holiday

[2]

ii)

a) Examples of Mortality Selection:

1. To allow for increase in the sum assured at any point in time, in particular in term insurance plan, without subjecting policyholder to any further underwriting;
2. To allow for increase in the term under term insurance plan without any further underwriting;

3. To accept online term insurance or government sponsored insurance plans only with a self-declaration of good health;

[2]

b) Examples of Financial Selection:

1. To offer guarantees under the unit funds and allowing the policyholder to switch between the funds without impacting the guarantee.
2. To offer highest guaranteed NAV even on surrender of the policy;

[2]

[6 Marks]
